

Medicine Reminder App - Design Document

1. Overview

The Medicine Reminder App is a mobile application developed using Android Studio with Java. It helps users manage their medication schedules efficiently by providing reminders, tracking progress, and allowing customization through user settings.

While designing the Medicine Reminder App, my main focus is to strengthen my skills in Android development using Java and Android Studio, as emphasized in the course exercises. I aim to learn how to design a functional and user-friendly application that serves a real-world purpose. Specifically, I want to improve my understanding of how to:

- Structure an Android app with clear separation of concerns between different classes (such as UI, database access, and logic),
- Design intuitive and accessible user interfaces,
- Integrate and manage a local SQLite database to store and query reminder data efficiently,
- Implement background tasks such as time-based notifications reliably.

When I designed my project, I thought my timetable would be:

- 24.2 - 2.3: UI design
- 3.3 - 30.3: Frontend pages
- 31.3 - 20.4: Backend and deployment
- 21.4 - 30.4: Documentation

However, due to academic pressures, my actual timetable was:

- 25.2: Design project
- 25.4 - 10.5: Research and programming
- 23.5 - 30.5: Debugging and Documentation

I also gave up the idea of including a backend in this project.

2. Features

2.1 Pill Reminders

- **Create New Reminder:**
 - Users can set reminders with a start date and end date.
 - Frequency options: Daily, Weekly, Monthly and Yearly, and Multiple times a day.
 - Validation checks: Ensure all required fields are filled before adding a reminder.
- **View and Manage Reminders:**
 - Reminders are sorted in chronological order.
 - Users can select different dates, and the interface will display reminders for the corresponding dates.
 - Users can mark a reminder as "done."
- **Notifications:**

- Timed alerts when a medication needs to be taken.

2.2 Progress Tracking

- Users can generate records (charts) of completed reminders.
- Provides daily, weekly, and monthly completion rate tracking.
- Displays visual representations of medication adherence.

2.3 Settings Menu

- Preferred language (Localization support for at least two languages)
- Progress Chart